

Lectures on Monte Carlo Theory

Chapter 5: Variance Reduction Techniques

Based on Leskelä & Vihola

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Motivation: Why Reduce Variance?

Crude Monte Carlo Estimator

For $I = \mathbb{E}[Y]$, the CMC estimator $\hat{Y}_R = \frac{1}{R} \sum_{i=1}^R Y_i$ has variance $\text{Var}(\hat{Y}_R) = \text{Var}(Y)/R$.

Key observations:

- To halve the error, we need *four times* as many samples.
- Reducing $\text{Var}(Y)$ by a factor k is equivalent to running k times more simulations.
- Variance reduction techniques exploit structure in the problem to achieve this without extra function evaluations.

Methods covered in this chapter:

- 1 Stratified sampling
- 2 Antithetic variates
- 3 Common random numbers
- 4 Ratios of expectations
- 5 Control variates
- 6 Conditional Monte Carlo
- 7 Importance sampling
- 8 Cross-entropy method

5.1.1 Stratified Sampling I

Setup

Partition the probability space Ω into m disjoint strata $A^1, \dots, A^m$ with $\mathbb{P}(Y \in A^j) = p_j > 0$, $\sum_{j=1}^m p_j = 1$.

Stratified Estimator

$$\hat{Y}_R^{\text{str}} = \sum_{j=1}^m p_j \hat{Y}_{R_j}^j = \frac{1}{m} \sum_{j=1}^m \hat{Y}_{R_j}^j \quad (\text{equal probabilities})$$

where $\hat{Y}_{R_j}^j = \frac{1}{R_j} \sum_{i=1}^{R_j} Y_i^j$ and $Y_i^j \sim (Y \mid Y \in A^j)$, independently across strata.

Symbols:

- R_j – number of replications allocated to stratum j
- $\sigma_j^2 = \text{Var}(Y \mid Y \in A^j)$ – within-stratum variance

5.1.1 Stratified Sampling II

Variance of Stratified Estimator

$$\text{Var}(\hat{Y}_R^{\text{str}}) = \sum_{j=1}^m \frac{p_j^2 \sigma_j^2}{R_j}$$

Optimal Allocation (Theorem 5.1.5)

Minimize variance subject to $\sum_{j=1}^m R_j = R$:

$$R_j = \frac{p_j \sigma_j}{\sum_{k=1}^m p_k \sigma_k} R$$

The resulting variance is:

$$\text{Var}(\hat{Y}_R^{\text{str,opt}}) = \frac{1}{R} \left(\sum_{j=1}^m p_j \sigma_j \right)^2$$

5.1.1 Stratified Sampling III

Proportional Allocation

$$R_j = p_j R \quad \Rightarrow \quad \text{Var}(\hat{Y}_R^{\text{prop}}) = \frac{1}{R} \sum_{j=1}^m p_j \sigma_j^2$$

5.1.1 Stratified Sampling IV

Variance Ordering

$$\text{Var}(\hat{Y}_R^{\text{opt}}) \leq \text{Var}(\hat{Y}_R^{\text{prop}}) \leq \text{Var}(\hat{Y}_R^{\text{CMC}})$$

Key insight: Stratified sampling with optimal allocation always reduces variance compared to CMC.

Example 5.1.8 – estimating π : Strata $A^j = (\frac{j-1}{m}, \frac{j}{m}]$ on $U_1 \sim \mathcal{U}[0, 1]$. $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$.
With $m = 20$ optimal strata: variance reduced by $\approx 9000\times$.

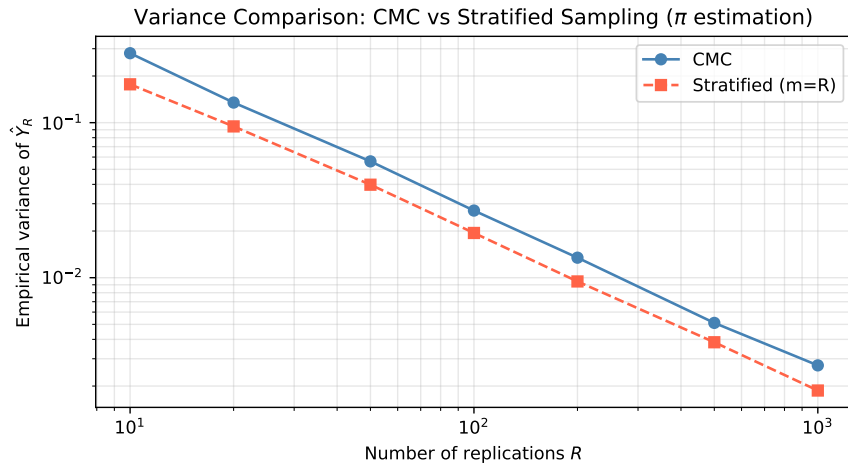
5.1.1 Stratified Sampling – Python Code

```
import numpy as np

def stratified_pi(m, R):
    """Estimate pi using stratified sampling on U1 with m strata."""
    R_j = R // m          # replications per stratum (equal allocation)
    estimates = []
    for j in range(m):
        # Sample U1 from stratum j: U1 in [(j)/m, (j+1)/m]
        U1 = (j + np.random.uniform(0, 1, R_j)) / m
        U2 = np.random.uniform(0, 1, R_j)
        Y = 4 * (U1**2 + U2**2 <= 1).astype(float)
        estimates.append(np.mean(Y))
    return np.mean(estimates)    # equal p_j = 1/m

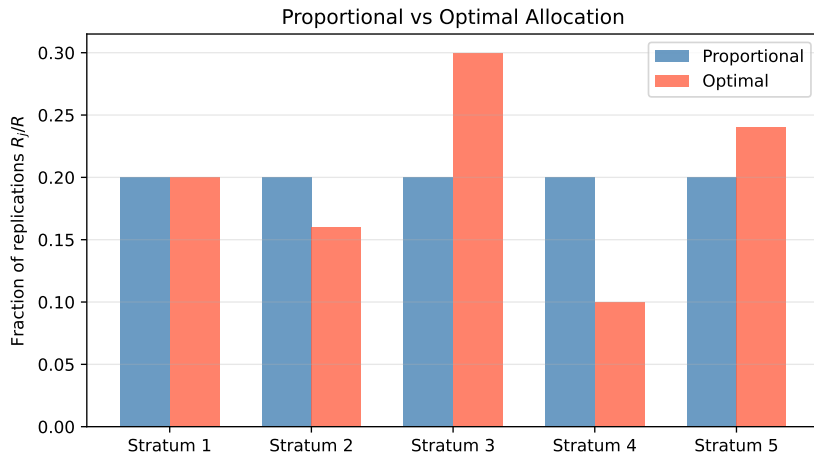
# Comparison
R = 100000
np.random.seed(42)
cmc = 4 * np.mean(np.random.uniform(0,1,(R,2)).__pow__(2).sum(axis=1) <= 1)
str_est = stratified_pi(m=100, R=R)
print(f"True pi = {np.pi:.5f}")
print(f"CMC:      {cmc:.5f}")
print(f"Stratified: {str_est:.5f}")
```


5.1.1 Stratified Sampling – Variance Comparison



Stratified sampling reduces variance significantly – the gap widens as m increases. Optimal allocation beats proportional, which beats CMC.

5.1.1 Optimal vs Proportional Allocation



Optimal allocation assigns *more* samples to strata with larger σ_j (within-stratum standard deviation), regardless of stratum probability p_j .

5.1.2 Antithetic Variates I

Idea

Generate pairs $(Y_1, Y'_1), (Y_2, Y'_2), \dots, (Y_R, Y'_R)$ such that $Y_j \stackrel{d}{=} Y'_j$ but $\text{Cov}(Y_j, Y'_j) < 0$. The *antithetic estimator* is:

$$\hat{Y}_R^{\text{anti}} = \frac{1}{2R} \sum_{j=1}^R (Y_j + Y'_j)$$

Variance Reduction

$$\text{Var}\left(\frac{Y + Y'}{2}\right) = \frac{\text{Var}(Y)}{2} (1 + \rho_{YY'})$$

where $\rho_{YY'} = \text{Corr}(Y, Y')$. Variance is reduced iff $\rho_{YY'} < 0$.

5.1.2 Antithetic Variates II

Standard Construction

If $Y = g(U_1, \dots, U_d)$ with $U_i \stackrel{\text{iid}}{\sim} \mathcal{U}[0, 1]$, set $Y' = g(1 - U_1, \dots, 1 - U_d)$.

Efficiency Condition (Proposition 5.1.11)

If g is *monotone* in each argument, then $\text{Cov}(Y, Y') \leq 0$, so antithetic variates reduce variance.

Example 5.1.13 – estimating $\mathbb{E}[e^U]$:

$$Y = e^U, \quad Y' = e^{1-U}, \quad U \sim \mathcal{U}[0, 1]$$
$$\text{Var}\left(\frac{Y + Y'}{2}\right) = \frac{3e^2 - e^4/2 - e^{-1}/2 - 2}{4} \approx \frac{0.00474}{1} \ll \text{Var}(Y) = 0.242$$

Reduction factor: $0.242/0.00474 \approx 51$.

5.1.2 Antithetic Variates – Python Code

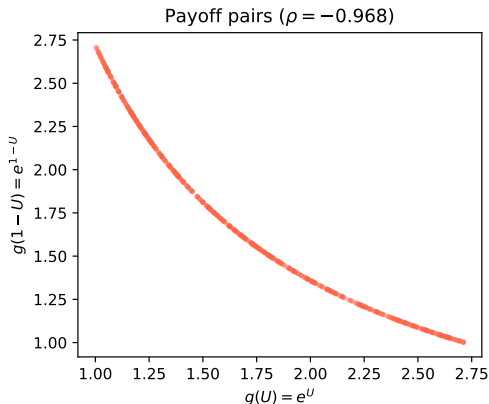
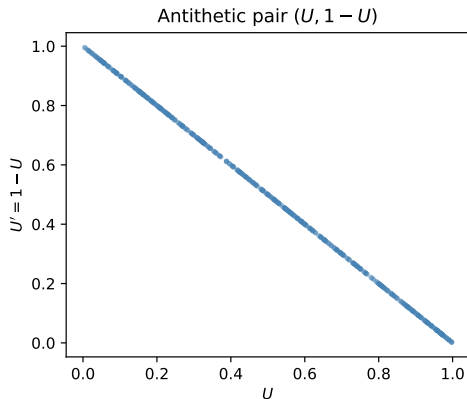
```
import numpy as np

def antithetic_pi(R):
    """Estimate pi using antithetic variates."""
    U1 = np.random.uniform(0, 1, R)
    U2 = np.random.uniform(0, 1, R)
    # Original samples
    Y = 4 * (U1**2 + U2**2 <= 1).astype(float)
    # Antithetic samples: replace U1 with 1-U1, U2 with 1-U2
    Yp = 4 * ((1-U1)**2 + (1-U2)**2 <= 1).astype(float)
    return np.mean((Y + Yp) / 2)

R = 100000; np.random.seed(42)
print(f"Antithetic pi estimate: {antithetic_pi(R):.6f}")
print(f"True pi: {np.pi:.6f}")

# Variance comparison
trials = [antithetic_pi(1000) for _ in range(2000)]
print(f"Var(antithetic): {np.var(trials):.6f}")
```

5.1.2 Antithetic Variates – Illustration



Left: The pair $(U, 1 - U)$ is perfectly negatively correlated. Right: The corresponding payoffs $g(U)$ and $g(1 - U)$ also show negative correlation $\rho \approx -1$, leading to dramatic variance reduction.

5.1.3 Common Random Numbers I

Goal: Comparing Two Systems

Estimate $\Delta = \mathbb{E}[Y_1] - \mathbb{E}[Y_2]$ where Y_1, Y_2 are outputs of two different system configurations.

Common Random Numbers (CRN)

Use the *same* random number stream to simulate both systems:

$$\hat{\Delta}_R^{\text{CRN}} = \frac{1}{R} \sum_{j=1}^R (Y_{1,j} - Y_{2,j})$$

where $(Y_{1,j}, Y_{2,j})$ are simulated with the same U_j .

5.1.3 Common Random Numbers II

Variance

$$\text{Var}(\hat{\Delta}_R^{\text{CRN}}) = \frac{\text{Var}(Y_1) + \text{Var}(Y_2) - 2\text{Cov}(Y_1, Y_2)}{R}$$

CRN reduces variance iff $\text{Cov}(Y_1, Y_2) > 0$, which holds when both outputs are monotone functions of the same input.

5.1.3 Common Random Numbers III

Example 5.1.20 – estimating π with CRN:

$$Y_1 = 8/(1 + U^2) - 4\sqrt{1 - U^2}, \quad Y_2 = \text{another estimator of } \pi$$
$$\hat{Y}_R^{\text{CRN}} = \frac{1}{R} \sum Y_1 \quad \text{with cross-linked random numbers}$$

Variance: 2.885×10^{-5} vs 2.697×10^{-4} for CMC.

Synchronisation: When systems have different structures, careful mapping of random numbers to the same logical variable is required (*synchronisation*).

5.1.4 Ratios of Expectations I

Problem

Estimate $I = \mathbb{E}[Y]/\mathbb{E}[Z]$ (ratio of two expectations).

Ratio Estimator

$$\hat{Y}_R^{\text{ratio}} = \frac{\hat{Y}_R}{\hat{Z}_R} = \frac{\frac{1}{R} \sum_{i=1}^R Y_i}{\frac{1}{R} \sum_{i=1}^R Z_i}$$

This is a *biased* estimator, but consistent.

Variance via Delta Method

$$\text{Var}(\hat{Y}_R^{\text{ratio}}) \approx \frac{1}{R(\mathbb{E}Z)^2} (\text{Var}(Y) + I^2 \text{Var}(Z) - 2I \text{Cov}(Y, Z))$$

Application: Self-normalised importance sampling uses a ratio estimator when the normalisation constant is unknown.

5.1.5 Control Variates I

Setup

We have a primary estimator Y of $I = \mathbb{E}[Y]$ and a *control variate* X with *known* $\mathbb{E}[X] = \mu_X$. The control variate estimator is:

$$\hat{Y}_R^{\text{cv}} = \hat{Y}_R^{\text{CMC}} + c(\hat{X}_R - \mu_X)$$

where c is a scalar coefficient and $\hat{X}_R = \frac{1}{R} \sum_{i=1}^R X_i$.

Symbols:

- c – control variate coefficient (scalar, to be chosen)
- $\mu_X = \mathbb{E}[X]$ – known mean of control variate
- $\rho = \text{Corr}(X, Y)$ – correlation between X and Y

5.1.5 Control Variates II

Optimal Coefficient and Variance

$$c^* = -\frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\text{Var}(\hat{Y}_R^{\text{cv}}) = \frac{\text{Var}(Y)}{R}(1 - \rho^2)$$

Variance reduction factor: $1/(1 - \rho^2)$.

Estimated Coefficient

In practice, c^* is unknown and estimated from R' pilot simulations:

$$\hat{c} = -\frac{\hat{S}_{XY}^2}{\hat{S}_X^2}$$

where $\hat{S}_{XY}^2$ is the sample covariance and $\hat{S}_X^2$ the sample variance.

5.1.5 Control Variates III

Example 5.1.24 – π estimation with control variate:

- $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$, $X = \mathbf{1}(U_1 + U_2 \leq 1)$, $\mu_X = 1/2$
- $\rho = \text{Corr}(X, Y) \approx 0.527$, reduction: $\approx 1.4\times$
- Using $X = \mathbf{1}(U_1 + U_2 > \sqrt{2})$: $1 - \rho^2 = 0.242$, reduction: $\approx 4\times$
- Using 4th-degree polynomial X : $1 - \rho^2 = 0.0000859848$, reduction: $\approx 11630\times!$

5.1.5 Control Variates – Python Code

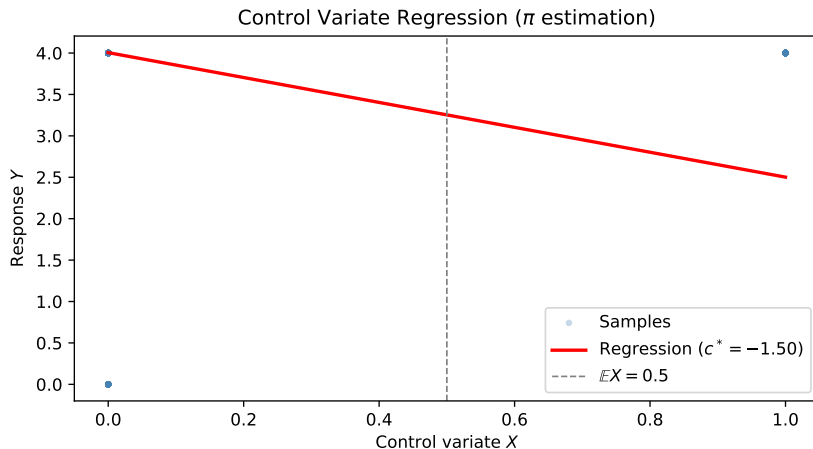
```
import numpy as np

def control_variate_pi(R, R_pilot=1000):
    """Estimate pi using a control variate  $X = 1(U_1+U_2 \leq 1)$ ,  $EX = 0.5$ ."""
    # Pilot simulation to estimate c
    U1p = np.random.uniform(0,1,R_pilot)
    U2p = np.random.uniform(0,1,R_pilot)
    Yp = 4*(U1p**2 + U2p**2 <= 1).astype(float)
    Xp = (U1p + U2p <= 1).astype(float)
    cov_XY = np.cov(Xp, Yp, ddof=1)
    c_hat = -cov_XY[0,1] / cov_XY[0,0]

    # Main simulation
    U1 = np.random.uniform(0,1,R); U2 = np.random.uniform(0,1,R)
    Y = 4*(U1**2 + U2**2 <= 1).astype(float)
    X = (U1 + U2 <= 1).astype(float)
    Y_cv = Y + c_hat * (X - 0.5)
    return np.mean(Y_cv), np.std(Y_cv) / np.sqrt(R)

np.random.seed(42)
est, se = control_variate_pi(R=100000)
print(f"CV estimate: {est:.6f} +/- {1.96*se:.6f}")
print(f"True pi:      {np.pi:.6f}")
```

5.1.5 Control Variates – Regression View



The control variate estimator fits a regression line through the (X, Y) scatter. The optimal coefficient $c^* = -\text{Cov}(X, Y)/\text{Var}(X)$ is the negative slope of this regression.

5.1.6 Conditional Monte Carlo (Rao-Blackwell) I

Conditional Estimator

Given a random variable X and $I = \mathbb{E}[Y] = \mathbb{E}[c(X)]$ where $c(x) = \mathbb{E}[Y \mid X = x]$:

$$\hat{Y}_R^{\text{cond}} = \frac{1}{R} \sum_{j=1}^R c(X_j)$$

where $X_1, \dots, X_R$ are i.i.d. copies of X .

Rao-Blackwell Theorem

$$\text{Var}(c(X)) \leq \text{Var}(Y)$$

by Jensen's inequality: $\mathbb{E}[Y^2 \mid X] \geq (\mathbb{E}[Y \mid X])^2$. Also:

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]) \geq \text{Var}(c(X)).$$

5.1.6 Conditional Monte Carlo (Rao-Blackwell) II

Example 5.1.29 – π estimation:

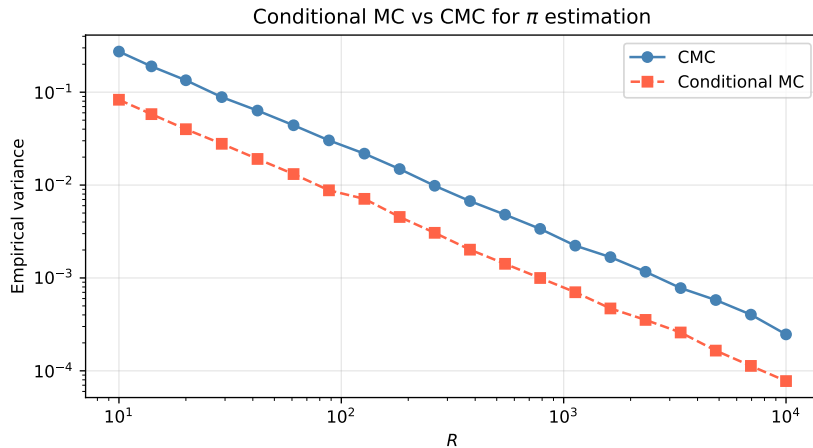
$$Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1), \quad X = U_1$$

$$c(x) = \mathbb{E}[Y \mid X = x] = 4\mathbb{P}(U_2 \leq \sqrt{1 - x^2}) = 4\sqrt{1 - x^2}$$

$$\hat{Y}_R^{\text{cond}} = \frac{4}{R} \sum_{i=1}^R \sqrt{1 - U_i^2}$$

- $\text{Var}(\hat{Y}_R^{\text{CMC}}) = 2.697/R$
- $\text{Var}(\hat{Y}_R^{\text{cond}}) = 0.797/R$
- Reduction factor: ≈ 3.38

5.1.6 Conditional MC – Variance Comparison



The conditional MC estimator consistently achieves lower variance. Both converge at rate $1/R$, but CMC has a higher constant.

5.1.7 Importance Sampling I

Change of Measure

Goal: estimate $I = \int_E k(x) f(x) dx = \mathbb{E}_f[k(X)]$. Let $\tilde{f}(x)$ be a *proposal density* (positive where $f > 0$). Then:

$$I = \int_E k(x) \frac{f(x)}{\tilde{f}(x)} \tilde{f}(x) dx = \mathbb{E}_{\tilde{f}} \left[k(\tilde{X}) \frac{f(\tilde{X})}{\tilde{f}(\tilde{X})} \right]$$

Importance Sampling Estimator

$$\hat{Y}_R^{\text{IS}} = \frac{1}{R} \sum_{j=1}^R k(\tilde{X}_j) \frac{f(\tilde{X}_j)}{\tilde{f}(\tilde{X}_j)}$$

where $\tilde{X}_1, \dots, \tilde{X}_R \stackrel{\text{iid}}{\sim} \tilde{f}$.

The ratio $L(x) = f(x)/\tilde{f}(x)$ is called the *likelihood ratio*.

5.1.7 Importance Sampling II

Variance of IS Estimator

$$\text{Var}(\hat{Y}_R^{\text{IS}}) = \frac{1}{R} \left[\int_E \left(k(x) \frac{f(x)}{\tilde{f}(x)} \right)^2 \tilde{f}(x) dx - I^2 \right]$$

Optimal Proposal Distribution

The zero-variance proposal is:

$$g^*(x) = \frac{k(x)f(x)}{\int_E k(x)f(x) dx} = \frac{k(x)f(x)}{I}$$

(impractical, since it requires knowing I ; but guides the choice of $\tilde{f}$).

5.1.7 Importance Sampling III

Proposition 5.1.33 – CLT for IS

$\hat{Y}_R^{\text{IS}}$ is unbiased and strongly consistent, and:

$$\sqrt{R}(\hat{Y}_R^{\text{IS}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2)$$

where $\varsigma^2 = \int_E (k(x)f(x)/\tilde{f}(x))^2 \tilde{f}(x) dx - I^2$.

5.1.7 Importance Sampling IV

Example 5.1.34 – $I = \mathbb{P}(X > 2)$ for Cauchy X :

- CMC: $\text{Var}(\hat{Y}_R^{\text{CMC}}) = I(1 - I)/R = 0.1258/R$
- IS with $\tilde{f}(x) = 2/x^2 \cdot \mathbf{1}(x > 2)$: $\text{Var}(\hat{Y}_R^{\text{IS}}) = 9.55 \times 10^{-5}/R$
- Reduction: $\approx 1317\times$

Example 5.1.36 – $I = \mathbb{P}(X > 4)$ for $X \sim \mathcal{N}(0, 1)$:

- $I \approx 3.167 \times 10^{-5}$
- CMC: $\text{Var} = I(1 - I)/R \approx 3.167 \times 10^{-5}/R$
- IS with $\tilde{f} = \mathcal{N}(4, 1)$: $\text{Var}(Y_1^{\text{IS}}) = 3.601 \times 10^{-8}$
- Reduction: $\approx 879\times$

5.1.7 Importance Sampling – Python Code

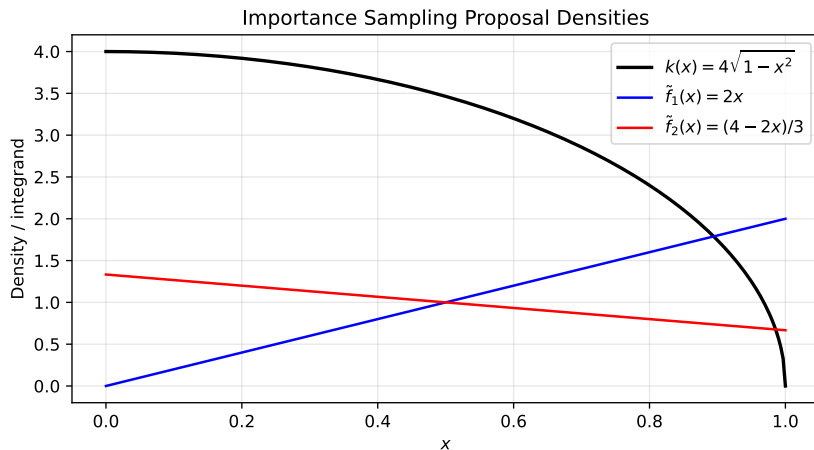
```
import numpy as np
from scipy.stats import norm

def IS_normal_tail(theta, R=100000):
    """Estimate  $P(Z > 4)$ ,  $Z \sim N(0,1)$  using IS with  $N(\theta, 1)$  proposal."""
    Z_tilde = np.random.normal(theta, 1, R)
    # Likelihood ratio:  $\phi(z; 0,1) / \phi(z; \theta, 1)$ 
    log_LR = -theta * Z_tilde + theta**2 / 2
    Y_IS = (Z_tilde > 4) * np.exp(log_LR)
    return np.mean(Y_IS), np.var(Y_IS) / R

np.random.seed(42)
true_I = 1 - norm.cdf(4)
print(f"True  $P(Z>4)$  = {true_I:.2e}")

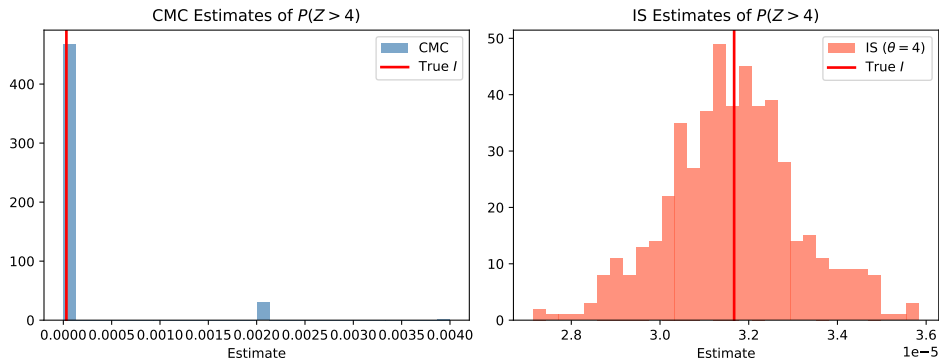
for theta in [0, 2, 4, 4.2256]:
    est, var = IS_normal_tail(theta)
    print(f"  theta={theta:.4f}: est={est:.2e}, var={var:.2e}, "
          f"reduction={true_I*(1-true_I)/var:.1f}x")
```

5.1.7 Importance Sampling – Proposal Distributions



The optimal proposal $\tilde{f} \propto k(x)f(x)$ should closely match the shape of the integrand. Among $\tilde{f}_1 = 2x$ and $\tilde{f}_2 = (4-2x)/3$, the one closer to $k(x)$ achieves lower variance.

5.1.7 IS for Rare Events: $\mathbb{P}(Z > 4)$



CMC estimates are mostly 0 (the event rarely occurs). IS with $\tilde{f} = \mathcal{N}(4, 1)$ concentrates samples in the rare-event region, giving consistent estimates.

Self-Normalised Importance Sampling I

When normalisation is unknown

If $f(x) = c \cdot f_0(x)$ and $\tilde{f}(x) = \tilde{c} \cdot \tilde{f}_0(x)$ with unknown constants, define $w(x) = f_0(x)/\tilde{f}_0(x)$ and:

$$I = \frac{\mathbb{E}_{\tilde{f}}[k(\tilde{X})w(\tilde{X})]}{\mathbb{E}_{\tilde{f}}[w(\tilde{X})]}$$

$$\hat{Y}_R = \frac{\sum_{i=1}^R k(\tilde{X}_i)w(\tilde{X}_i)}{\sum_{i=1}^R w(\tilde{X}_i)}$$

This is the *self-normalised* (or ratio) IS estimator.

Estimated Variance (Delta Method)

$$\widehat{\text{Var}}(\hat{Y}_R) = \sum_{i=1}^R (w'(\tilde{X}_i))^2 (k(\tilde{X}_i) - \hat{Y}_R)^2$$

where $w'(\tilde{X}_i) = w(\tilde{X}_i) / \sum_j w(\tilde{X}_j)$.

Self-Normalised Importance Sampling III

Likelihood Ratio (Proposition 5.1.42)

Under two probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$ with $X_1, X_2, \dots$ i.i.d. with densities f and $\tilde{f}$:

$$L_n = \prod_{j=1}^n \frac{f(X_j)}{\tilde{f}(X_j)}$$

then $\mathbb{P}(d\omega) = L_n(\omega)\tilde{\mathbb{P}}(d\omega)$, and for any integrable Y : $I = \mathbb{E}_f Y = \tilde{\mathbb{E}}[Y L_n]$.

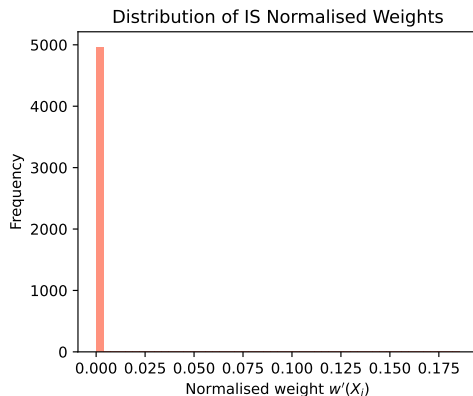
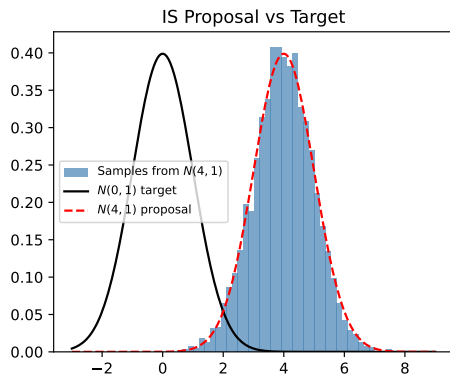
Exponential Change of Measure (Rare Events)

For X_i with p.d.f. $f(x)$, define $f_\theta(x) = e^{\theta x} f(x) / M(\theta)$ where $M(\theta) = \mathbb{E}[e^{\theta X}]$. Then:

$$I(n) = \mathbb{P}(S_n > n(\mu + \varepsilon)) = \mathbb{E}_\theta \left[\mathbf{1}_{S_n > n(\mu + \varepsilon)} e^{-\theta S_n + n\kappa(\theta)} \right]$$

Choose $\theta = \theta_0$ such that $\kappa'(\theta_0) = \mu + \varepsilon$ for the logarithmically efficient estimator.

5.1.7 Self-Normalised IS Weights



Using a proposal $N(4, 1)$ concentrates samples in the rare region. The resulting normalised weights $w'(X_i)$ have low variance, ensuring a stable estimator.

5.1.8 The Cross-Entropy Method I

Motivation

How to choose the best proposal $\tilde{f}_\theta$ from a parametric family $\{\tilde{f}_\theta : \theta \in \Theta\}$?

KL Divergence and Optimal θ^*

Minimise the Kullback-Leibler divergence from $g^*(x) = k(x)f(x)/I$ to $\tilde{f}_\theta$:

$$\theta^* = \arg \max_{\theta} \int k(\mathbf{x}) \log \tilde{f}_\theta(\mathbf{x}) f(\mathbf{x}) d\mathbf{x} = \arg \max_{\theta} \mathbb{E}_f \left[k(\mathbf{X}) \log \tilde{f}_\theta(\mathbf{X}) \right]$$

Estimated from M samples as:

$$\hat{\theta}^* = \arg \max_{\theta} \frac{1}{M} \sum_{i=1}^M k(\mathbf{X}_i) \log \tilde{f}_\theta(\mathbf{X}_i)$$

5.1.8 The Cross-Entropy Method II

CE Method for Rare Event Simulation

For $I = \mathbb{P}_f(h(\mathbf{X}) \geq \gamma)$, iteratively:

- ① Update level: $\gamma_t = (1 - \rho)$ -quantile of $h(\mathbf{X}_i)$
- ② Update parameter using likelihood ratio $L(\mathbf{X}, f, f_{\theta'_{t-1}})$:

$$\theta'_t = \arg \max_{\theta} \frac{1}{M} \sum_{i=1}^M \mathbf{1}(h(\mathbf{X}_i) \geq \gamma_t) L(\mathbf{X}_i, f, f_{\theta'_{t-1}}) \log f_{\theta}(\mathbf{X}_i)$$

- ③ Apply *smoothing*: $\theta'_t := \alpha \theta'_t + (1 - \alpha) \theta'_{t-1}$
- ④ Stop when $\gamma_t \geq \gamma$; final IS estimator:

$$\hat{Y}_R^{\text{CE}} = \frac{1}{R} \sum_{i=1}^R \mathbf{1}(h(\mathbf{X}_i) > \gamma) L(\mathbf{X}_i, f, f_{\theta_T})$$

5.1.8 The Cross-Entropy Method III

CE Update for Normal Family $\tilde{f}_\theta = \mathcal{N}(\theta, \Sigma)$

The optimal parameter update is a weighted mean:

$$\theta'_t = \frac{\sum_{i=1}^M \mathbf{1}(h(\mathbf{X}_i) \geq \gamma_t) w_{i,t-1} \mathbf{X}_i}{\sum_{i=1}^M \mathbf{1}(h(\mathbf{X}_i) \geq \gamma_t) w_{i,t-1}}$$

where $w_{i,t-1} = L(\mathbf{X}_i, f, f_{\theta'_{t-1}})$.

Example 5.1.47 – $\mathbb{P}(X > 4)$ for $X \sim \mathcal{N}(0, 1)$:

- CE with $\{\mathcal{N}(\theta, 1)\}$ family: $\theta^* = 4.2256$
- $\text{Var}(\hat{Y}_R^{\text{IS}})$ at $\theta = 4$: 3.601×10^{-8}
- At CE $\theta^* = 4.2256$: 4.516×10^{-9} (better by $8\times$)
- Algorithm converged in 46 steps (Example 5.1.54)

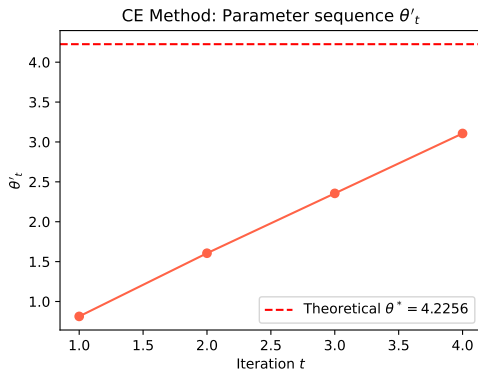
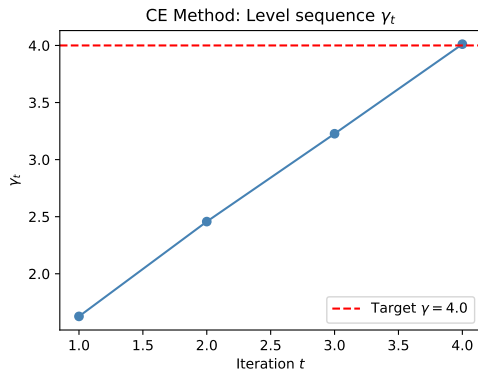
5.1.8 Cross-Entropy Method – Python Code

```
import numpy as np

def CE_normal_1D(M=10000, rho=0.05, alpha=0.4, gamma_target=4.0):
    """Cross-entropy method to estimate  $P(Z > 4)$ ,  $Z \sim N(0,1)$ ."""
    theta = 0.0
    for t in range(100):
        X = np.random.normal(theta, 1, M)
        # Likelihood ratio to original  $N(0,1)$ 
        w = np.exp(-theta * X + theta**2 / 2)
        # Update level (1-rho quantile)
        gamma_t = np.quantile(X, 1 - rho)
        # Update theta: weighted mean of X satisfying  $h(X) \geq \gamma_t$ 
        mask = X >= gamma_t
        if mask.sum() > 0:
            new_theta = np.sum(w[mask] * X[mask]) / np.sum(w[mask])
            theta = alpha * new_theta + (1 - alpha) * theta
        print(f"t={t+1}: gamma={gamma_t:.3f}, theta={theta:.4f}")
        if gamma_t >= gamma_target:
            break
    return theta

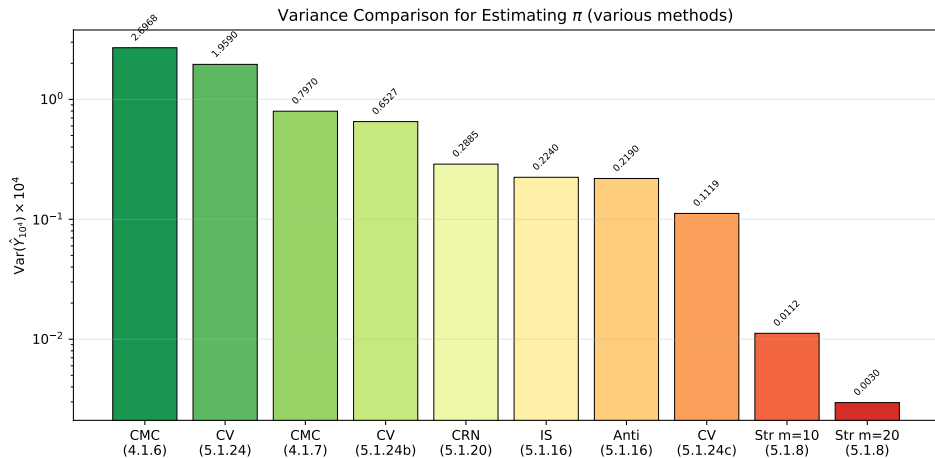
theta_CE = CE_normal_1D()
print(f"\nCE theta* = {theta_CE:.4f} (theory: 4.2256)")
```

5.1.8 Cross-Entropy Convergence



Left: Level sequence γ_t converges to target $\gamma = 4$. Right: Parameter θ'_t converges to theoretical optimum $\theta^* = 4.2256$.

5.1.9 Summary: Estimating π – Variance Comparison



5.1.9 Summary Table (from Table 5.11)

Method	Description	$\text{Var}(\hat{Y}_{10^4}) \times 10^4$	Reduction
CMC	$Y = 4\mathbf{1}(U_1^2 + U_2^2 \leq 1)$	2.697	1×
CV	$X = \mathbf{1}(U_1 + U_2 \leq 1)$	1.959	1.4×
CMC	$Y = 4\sqrt{1 - U^2}$	0.797	3.4×
CV	$X = \mathbf{1}(U_1 + U_2 \geq \sqrt{2})$	0.653	4.1×
CRN	$Y = 8/(1 + U^2) - 4\sqrt{1 - U^2}$	0.289	9.3×
IS	$\tilde{f} = (4 - 2x)/3$	0.224	12×
Anti	$Y_1 = 4\sqrt{1 - U^2}, Y_2 = 4\sqrt{1 - (1 - U)^2}$	0.219	12×
CV	poly. X , 4th degree	0.112	24×
Str	$m = 10$ strata, optimal	0.0112	240×
Str	$m = 20$ strata, optimal	0.00296	911×

5.2 Finance and Actuarial Science – Overview

Motivation

Classical financial mathematics is built on *geometric Brownian motion* (GBM). Options pricing requires computing expectations of functionals of GBM. Variance reduction techniques can dramatically reduce the computational cost.

Topics covered:

- 1 Brownian motion: definition, simulation, bisection method
- 2 Financial contracts: European/Asian options, Black-Scholes
- 3 Stratified sampling for option pricing
- 4 Control variates for options
- 5 Antithetic variates for options
- 6 Importance sampling for options

5.2.1 Brownian Motion and Related Processes I

Definition 5.2.1 – Standard Brownian Motion

A stochastic process $(B(t) : t \in [0, T])$ is a *standard Brownian motion* if:

- (i) $B(0) = 0$
- (ii) Increments $B(t_1), B(t_2) - B(t_1), \dots, B(t_n) - B(t_{n-1})$ are independent for all $0 \leq t_1 < \dots < t_n \leq T$
- (iii) $B(t_2) - B(t_1) \sim \mathcal{N}(0, t_2 - t_1)$ for $0 \leq t_1 \leq t_2$
- (iv) $t \mapsto B(t)$ is continuous a.s.

Proposition 5.2.2 – Characterisation

A process with a.s. continuous trajectories is a standard BM iff:

$$\text{Cov}(B(s), B(t)) = \min(s, t), \quad \text{for all } 0 \leq s \leq t \leq T.$$

5.2.1 Brownian Motion and Related Processes II

Simulation: Algorithm 36

On grid $t_i = i\Delta$, $\Delta = 1/N$, generate $Z_1, \dots, Z_N \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$:

$$B(i\Delta) = \sum_{k=1}^i \sqrt{\Delta} Z_k$$

Brownian Motion with Drift

$$X(t) = \sigma B(t) + \mu t, \quad \mu \in \mathbb{R}, \sigma > 0$$

5.2.1 Brownian Motion and Related Processes III

Geometric Brownian Motion (GBM)

$$S(t) = S(0) \exp\left(\left(r - \frac{\sigma^2}{2}\right)t + \sigma B(t)\right)$$

where r is the risk-free rate and σ the volatility.

5.2.1 Brownian Motion and Related Processes IV

Brownian Bridge (Bisection)

The *Brownian bridge* is BM conditioned on $B(1) = 0$. Given $B(0) = 0$ and $B(1) = y$, the midpoint $B(1/2)$ satisfies:

$$(B(1/2) \mid B(1) = y) \sim \mathcal{N}(y/2, 1/4)$$

This allows the *bisection algorithm* (Algorithm 38) that generates BM in order: $B(1), B(1/2), B(1/4), B(3/4), \dots$ Useful for stratified sampling of BM.

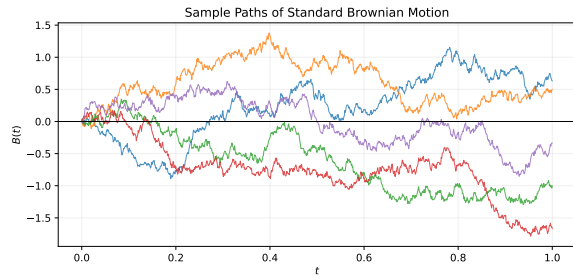
5.2.1 Brownian Motion – Python Code

```
import numpy as np
import matplotlib.pyplot as plt

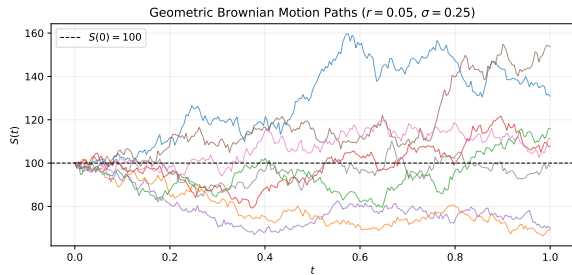
def sim_brownian(N=1000, T=1.0, n_paths=5):
    """Simulate Brownian motion on [0,T] with N steps."""
    dt = T / N
    t = np.linspace(0, T, N+1)
    paths = np.zeros((n_paths, N+1))
    for i in range(n_paths):
        Z = np.random.standard_normal(N)
        paths[i, 1:] = np.cumsum(np.sqrt(dt) * Z)
    return t, paths

def sim_gbm(S0=100, r=0.05, sigma=0.25, N=252, n_paths=5):
    """Simulate Geometric Brownian Motion (risk-neutral)."""
    dt = 1.0 / N
    mu = r - sigma**2 / 2
    t = np.linspace(0, 1, N+1)
    paths = np.zeros((n_paths, N+1))
    paths[:, 0] = S0
    for i in range(n_paths):
        Z = np.random.standard_normal(N)
        log_ret = mu * dt + sigma * np.sqrt(dt) * Z
        paths[i, 1:] = S0 * np.exp(np.cumsum(log_ret))
    return t, paths
```

5.2.1 Brownian Motion Sample Paths



Standard BM $B(t)$



GBM $S(t) = S(0) \exp(\mu^*t + \sigma B(t))$

5.2.2 Financial Contracts and Black-Scholes I

Common option types (under GBM(r, σ)):

- **European call:** $f(S(1)) = (S(1) - K)_+$
- **European put:** $f(S(1)) = (K - S(1))_+$
- **Asian call:** $f = \left(\frac{1}{n} \sum_{j=1}^n S(j/n) - K \right)_+$
- **Up-and-out call:** $(S(1) - K)_+ \cdot \mathbf{1}(S(t) < b \ \forall t)$
- **European basket call:** $\left(\sum_{j=1}^d w_j S_j(1) - K \right)_+$

Option Price under Risk-Neutral Measure

$$I = e^{-r} \mathbb{E}[f(S(t_1), \dots, S(t_n))]$$

where $S(t) = S(0) \exp(\mu^* t + \sigma B(t))$, $\mu^* = r - \sigma^2/2$.

5.2.2 Financial Contracts and Black-Scholes II

Black-Scholes Formula (European Call)

$$\mathbb{E}(S(1) - K)_+ = S(0)\Phi(d_1) - Ke^{-r}\Phi(d_2)$$

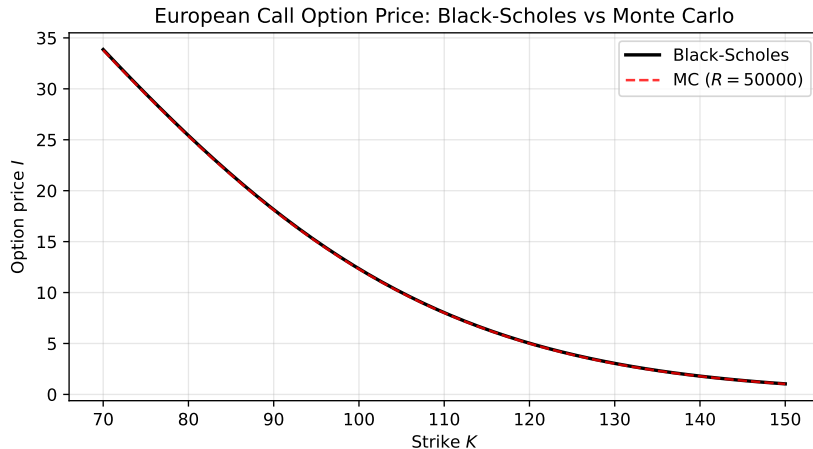
$$d_1 = \frac{1}{\sigma} \left[\log \left(\frac{S(0)}{K} \right) + r + \frac{\sigma^2}{2} \right], \quad d_2 = d_1 - \sigma$$

where Φ is the standard normal CDF.

Example parameters (used throughout):

- $r = 0.05$, $\sigma = 0.25$, $S(0) = 100$, $K = 100$
- $\mu^* = r - \sigma^2/2 = -0.01875$
- Exact Black-Scholes price $I \approx 12.336$

5.2.2 Black-Scholes vs Monte Carlo



Black-Scholes formula gives exact option prices; Monte Carlo approximates them with controlled statistical error.

5.2.3 Stratified Sampling for Finance I

Example 5.2.5 – European Call Option, m strata:

Method

Stratify on $B(1) \sim \mathcal{N}(0, 1)$: strata $A_j = (a_{j-1}, a_j]$ where $a_j = \Phi^{-1}(j/m)$.

- Proportional allocation: $R_j = R/m$
- Optimal allocation: $R_j \propto \hat{\sigma}_j$ (from pilot)

Results (Table 5.13) with $R = 10^5$, $S(0) = 100$, $r = 0.05$, $\sigma = 0.25$, $K = 100$:

Strata m	CMC	5	50	200
Estimate	12.280	12.351	12.338	12.336
Error b	0.115	0.043	0.025	0.002
Var reduction	–	7.2	21.0	3778

5.2.3 Stratified Sampling for Finance II

Example 5.2.8 – Asian Call Option with Stratified BM: Two methods for stratifying:

- 1 **Method 1:** Stratify on $(B(1/n), \dots, B(n/n))$ using multivariate normal strata (concentric ellipses via χ^2 distribution)
- 2 **Method 2:** Stratify Z_{2^k} and $Z_{2^{k-1}}$ in bisection algorithm; resulting in m^2 strata for $(B(1/2), B(1))$

Results with $K = 100$: $m = 200$ strata reduces variance by factor $84\times$.

5.2.4 Control Variates for Options Pricing I

Example 5.2.9 – Asian Call Option:

$$I = e^{-r} \mathbb{E}(A - K)_+, \quad A = \frac{1}{n} \sum_{j=1}^n S(j/n)$$

Control Variate: Geometric Mean

The geometric mean $G = \left(\prod_{j=1}^n S(j/n) \right)^{1/n}$ follows a log-normal distribution with exact analytical mean:

$$G = S(0) \exp[\mu_N + \sigma_N Z]$$

where $\mu_N = \mu^*(n+1)/(2n)$, $\sigma_N^2 = \sigma^2(n+1)(2n+1)/(6n^2)$, and $Z \sim \mathcal{N}(0, 1)$.

Control variate: $W = e^{-r} \mathbb{E}(G - K)_+$ (computed via Black-Scholes).

5.2.4 Control Variates for Options Pricing II

Results (Table 5.17) with $R = 10^4$, $n = 10$, $K = 100$, $r = 0.05$:

K	CMC error	CV error	Reduction	Anti error
150	2.45	0.374	$\approx 40\times$	0.0153
120	10.5	0.586	$\approx 320\times$	0.0943
100	21.5	0.731	$\approx 870\times$	2.10
50	30.7	0.998	–	0.302

The CV estimator dramatically reduces error for out-of-the-money options.

5.2.5 Antithetic Variates for Options I

Example 5.2.10 – Asian Call with Antithetic Variates:

Construction

Generate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, compute original path $B(k/n) = n^{-1/2} \sum_{j=1}^k Z_j$, and antithetic path $B'(k/n) = n^{-1/2} \sum_{j=1}^k (-Z_j)$:

$$\hat{Y}_R^{\text{anti}} = \frac{1}{2R} \sum_{j=1}^R [e^{-r}(A_j - K)_+ + e^{-r}(A'_j - K)_+]$$

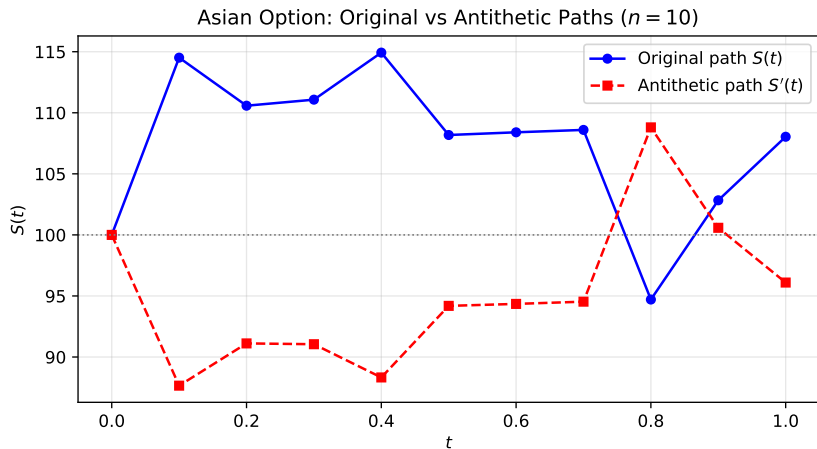
Key insight: $S(t)$ is a monotone increasing function of $B(t)$, and A is a monotone function of S , so antithetic variates guarantee $\text{Cov}(Y, Y') < 0$.

5.2.5 Antithetic Variates for Options II

Comparison with CMC and CV (Table 5.17):

- For $K = 150$: antithetic error = 0.0153 vs CMC error = 2.45 (reduction $\approx 25\,000\times$ in variance!)
- For $K = 120$: antithetic error = 0.0943 vs CMC error = 10.5
- Antithetic outperforms CV for deep out-of-the-money options

5.2.5 Antithetic Variates – Path Illustration



The antithetic path $S'(t) = S(0) \exp(\mu^* t - \sigma B(t))$ mirrors the original. When $S(t)$ tends up, $S'(t)$ tends down, creating negative correlation in payoffs.

5.2.6 Importance Sampling for Options I

Example 5.2.11 – European Call, IS with $\tilde{f} = \mathcal{N}(x^*, 1)$:

IS Estimator

Rewrite: $I = e^{-r} \int_{-\infty}^{\infty} (S(0)e^{\mu^* + \sigma z} - K)_+ \phi_{0,1}(z) dz$. Use proposal $\phi_{x^*,1}(z)$. IS estimator:

$$\hat{Y}_R^{\text{IS}} = \frac{1}{R} \sum_{i=1}^R e^{-r} (S(0)e^{\mu^* + \sigma \tilde{Z}_i} - K)_+ e^{-x^* \tilde{Z}_i + (x^*)^2/2}$$

where $\tilde{Z}_i \sim \mathcal{N}(x^*, 1)$.

5.2.6 Importance Sampling for Options II

Two Methods for Choosing x^*

- **Heuristic:** $x^* = \arg \sup_z h(z)$ where $h(z) = (S(0)e^{\mu^* + \sigma z} - K)_+ \phi_{0,1}(z) \Rightarrow$ solve $S(0)\sigma e^{\mu^* + \sigma z} - z(S(0)e^{\mu^* + \sigma z} - K) = 0$
- **CE method:** $\hat{x}_{\text{CE}}^* = \frac{\sum_i Z_i (S(0)e^{\mu^* + \sigma Z_i} - K)_+}{\sum_i (S(0)e^{\mu^* + \sigma Z_i} - K)_+}$

5.2.6 Importance Sampling for Options III

Results (Table 5.18) with $S(0) = K = 100$, $r = 0.05$, $\sigma = 0.25$:

	CMC	IS ($x^* = 1.033$)	IS CE ($x^* = 1.232$)
Estimate	12.391	12.302	12.306
Error b	0.387	0.125	0.120
Var reduction	–	9.65	10.46

Both IS methods achieve $\approx 10\times$ variance reduction. CE gives slightly better results.

5.2.6 IS for European Call – Python Code

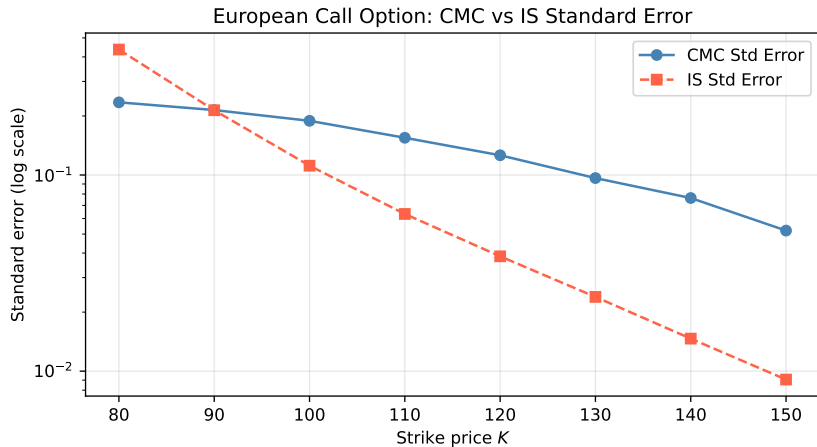
```
import numpy as np
from scipy.optimize import brentq
from scipy.stats import norm

def european_call_IS(S0=100, K=100, r=0.05, sigma=0.25, R=100000):
    """IS estimator for European call using CE-optimal x*."""
    mu_star = r - sigma**2 / 2
    # Step 1: Estimate x* via cross-entropy (closed form for normal)
    M = 10000
    Z = np.random.standard_normal(M)
    S1 = S0 * np.exp(mu_star + sigma * Z)
    payoff = np.maximum(S1 - K, 0)
    x_star = np.sum(Z * payoff) / np.sum(payoff)

    # Step 2: IS estimation
    Z_tilde = np.random.normal(x_star, 1, R)
    S1t = S0 * np.exp(mu_star + sigma * Z_tilde)
    log_LR = -x_star * Z_tilde + x_star**2 / 2
    Y_IS = np.exp(-r) * np.maximum(S1t - K, 0) * np.exp(log_LR)
    est = np.mean(Y_IS)
    se = np.std(Y_IS) / np.sqrt(R)
    return est, se, x_star

est, se, xs = european_call_IS()
print(f"IS estimate: {est:.4f} +/- {1.96*se:.4f}")
print(f"CE x*: {xs:.4f}")
```

5.2.6 IS for European Call – Standard Error Comparison



IS achieves substantially lower standard error for deep out-of-the-money options (large K), where CMC fails because the event is rare.

5.3 Bibliographical Notes

Key references:

- **Stratified sampling and variance reduction fundamentals:** Rubinstein & Kroese [186]; Asmussen & Glynn [8]
- **Cross-entropy method:** Rubinstein [184] (original); Rubinstein & Kroese [186] (comprehensive)
- **Practical tutorial on CE:** de Boer [31]
- **Brownian motion bisection method:** Asmussen & Glynn [8]; Knuth [106]
- **Variance reduction for quantile/VaR estimation:** Chu & Nakayama [25], Dong & Nakayama [39], Asmussen [6]
- **Finance:** Black & Scholes (1973); Hull [96]

5.4 Selected Exercises I

Exercise 5.T.1 (Stratified Sampling) Suppose we estimate $I = \mathbb{E}Y$ with strata $A^1, \dots, A^m$, $p_j = \mathbb{P}(Y \in A^j)$, cost per replication in stratum j is t_j . Total budget C .

- (a) Show $\sqrt{C}(\hat{Y}_C^{\text{str}} - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \zeta^2)$ where $\zeta^2 = \sum_{j=1}^m \frac{p_j^2 \sigma_j^2}{\gamma_j} \sum_{k=1}^m t_k \gamma_k$.
- (b) Optimal allocation: $R_j \propto p_j \sigma_j / \sqrt{t_j}$.

Exercise 5.T.8 (Optimal IS Distribution) Show that $g^*(x) = k(x)f(x)/I$ is the solution to a variational problem: $\arg \min_g \text{Var}_g(k(X)f(X)/g(X))$.

Exercise 5.T.25 (CE Optimal Parameter) For $I = \mathbb{P}(X > \gamma)$, $X \sim \mathcal{N}(0, 1)$, show that the CE-optimal mean parameter for $\{\mathcal{N}(\theta, 1)\}$ family is:

$$\theta^* = \frac{\sqrt{2}e^{-\gamma^2/2}}{\sqrt{\pi}(1 - \text{erf}(\gamma/\sqrt{2}))}$$

Chapter 5 Summary

Core Message

All variance reduction techniques aim to reduce $\text{Var}(Y)$, equivalent to obtaining more information per simulation run.

Method	Key idea	When to use
Stratified	Partition input space	Known input structure
Antithetic	Negative correlation pairs	Monotone functions
CRN	Shared randomness	Comparing two systems
Control variates	Regression correction	Correlated auxiliary known
Conditional MC	Rao-Blackwell	Can compute conditional expectation
Importance sampling	Change of measure	Heavy tails / rare events
Cross-entropy	Adaptive IS	Choose optimal IS distribution

Finance applications: Stratification + CV + antithetics + IS can each reduce variance by orders of magnitude for option pricing.